Human Biology GCE O Level 2022 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0565 Human Biology 1, June 2022, Ordinary Level, General Certificate of Education Board). Answer ALL 50 questions; all questions carry equal marks. Each question has four suggested answers A, B, C and D; select only one answer. Non-programmable calculators are allowed.

Multiple choice questions - choose the one best answer, A, B, C or D. Questions 12-15 consist of two statements.

1. How are humans different from other mammals?

a) Humans are heterodont
b) Humans are viviparous
c) Humans have pinnae
d) Humans have erect postures

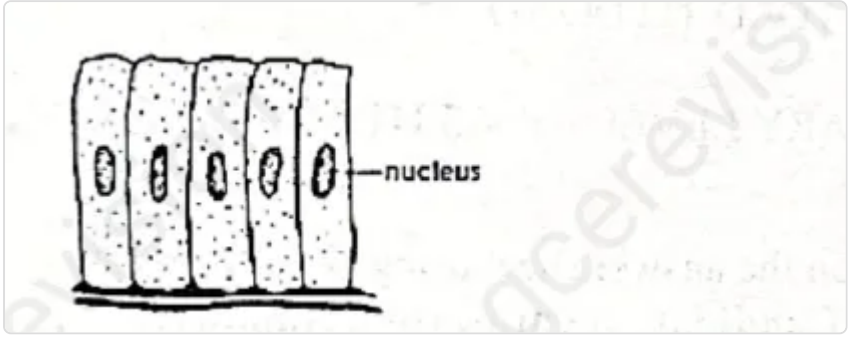
2. What is active transport?

a) Movement of substances from an area of low concentration to that of high concentration
b) Movement of solutes from an area of high concentration to that of low concentration
c) Movement of substances from an area of high concentration to that of low concentration
d) Movement of water from an area of low water potential to that of high water potential.

3. A cell organelle that plays a role in cell division is:

a) Lysosome
b) Ribosome
c) Nucleus
d) Golgi bodies

4. Identify the epithelium shown below.



(Courtsey of Roland Soper, GCSE, Human Biology).

a) Cuboidal
b) Columnar
c) Squamous
d) Transitional

5. A cell with 8 chromosomes undergoes mitosis. How many chromosomes will be found in each daughter cell?

a) 16
b) 4
c) 32
d) 8

6. A natural fibre which is good for the production of sport wear is:

a) cotton
b) wool
c) Silk
d) linen

7. Which type of muscle makes up the walls of blood vessels?

a) Cardiac muscle
b) Skeletal muscle
c) Smooth muscle
d) Striated muscle

8. Which of the following reduces friction at the joint?

a) Tendon and ligaments
b) Cartilage and synovial fluid
c) Ligaments and cartilage
d) Synovial fluid and tendons

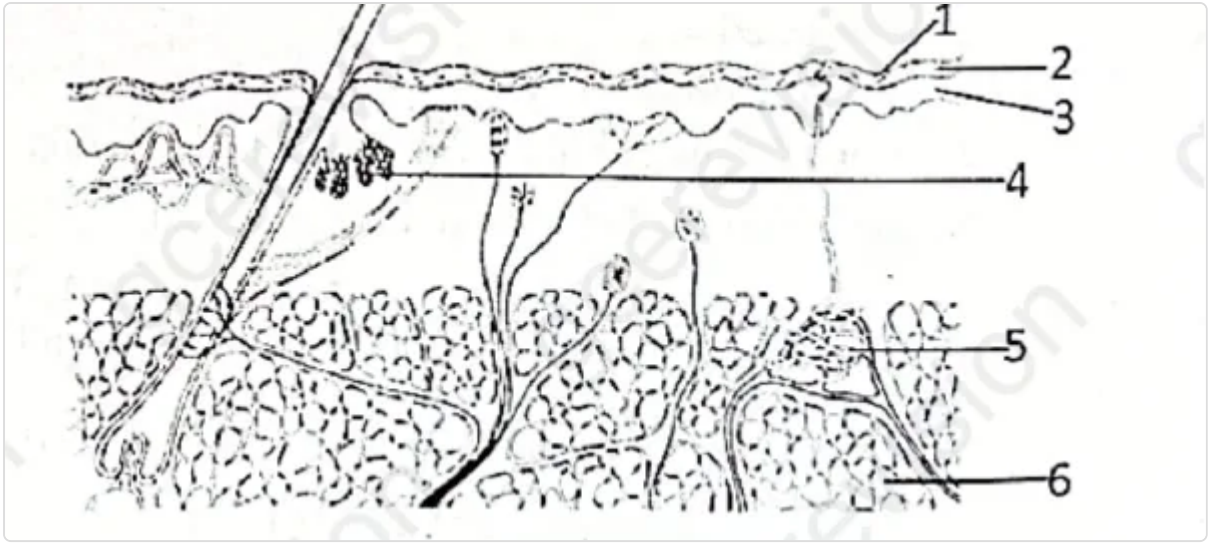
9. The chamber of the ear filled with fluid is:

a) Inner ear
b) Outer ear
c) Middle ear
d) Eustachian tube

10. The ions responsible for the conduction of nerve impulses along the neurone are:

a) Sodium and calcium
b) Potassium and calcium
c) Potassium and magnesium
d) Sodium and potassium

11. Below is the diagram of the skin.



(Courtesy of DG Macheans, Introduction to Biology).

Identify the part that regulates body temperature.

a) 5
b) 4
c) 6
d) 2

12. Questions 12-15 consist of two statements. Read through the statements and then choose A, B, C or D.

First statement	Second statement
The afferent arteriole which brings blood to the glomerulus is wider in diameter than the efferent arteriole which takes blood away	The blood in the glomerulus is under high pressure for ultrafiltration

a) If both statements are true and the second explains the first
b) If both statements are true but the second does not explain the first
c) If the first statement is false and the second is true
d) If both statements are false

13. Questions 12-15 consist of two statements. Read through the statements and then choose A, B, C or D.

First statement	Second statement
The visible traits of an organism constitute its genotype	Albinism is a sex-linked recessive trait in humans

a) If both statements are true and the second explains the first
b) If both statements are true but the second does not explain the first
c) If the first statement is false and the second is true
d) If both statements are false

14. Questions 12-15 consist of two statements. Read through the statements and then choose A, B, C or D.

First statement	Second statement
Photosynthesis uses O_2 and produces CO_2	Autotrophs constitute the first trophic level of a food chain

a) If both statements are true and the second explains the first
b) If both statements are true but the second does not explain the first
c) If the first statement is false and the second is true
d) If both statements are false

15. Questions 12-15 consist of two statements. Read through the statements and then choose A, B, C or D.

First statement	Second statement
The presence of protein in a food sample will give a red precipitate	NaOH and copper II sulphate are used to test for the presence of proteins in a food sample

a) If both statements are true and the second explains the first
b) If both statements are true but the second does not explain the first
c) If the first statement is false and the second is true
d) If both statements are false

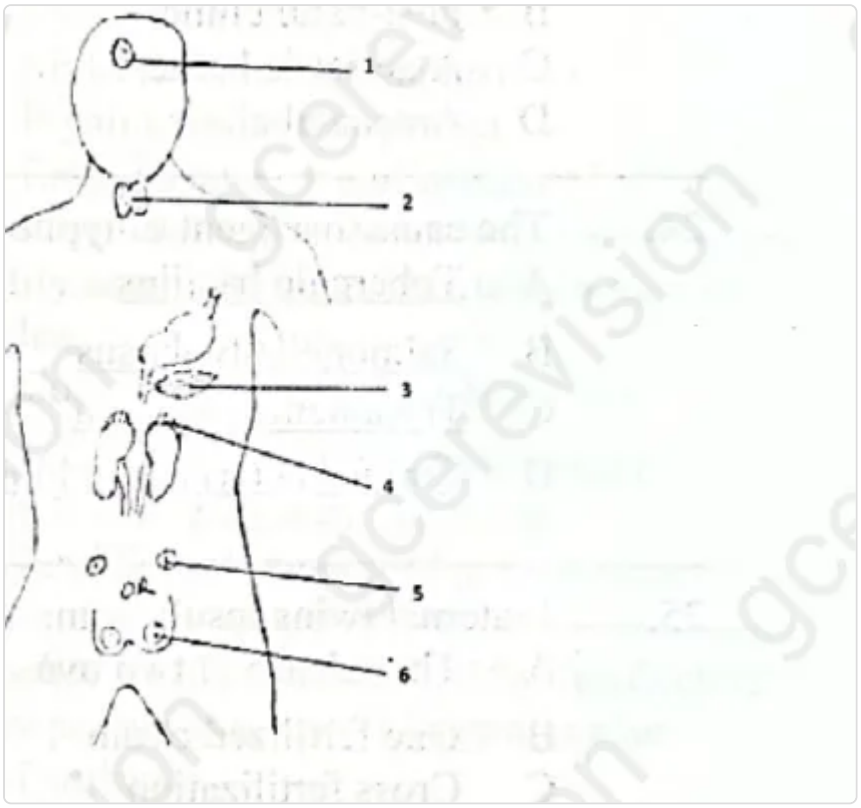
16. Cells that play a role in bone formation are:

a) Osteoblasts
b) Osteoclasts
c) Osteocytes
d) Chondroblasts

17. When alleles express themselves equally in the phenotype, the condition is referred to as:

a) Sex linkage
b) Codominance
c) Incomplete codominance
d) Dominance

18. Identify the endocrine gland below which secretes a hormone that regulates blood sugar level.



(Adapted from DG Macheans, Introduction to Biology).

a) 3
b) 1
c) 4
d) 2

19. Which of these blood groups is a universal recipient?

a) Group A
b) Group B
c) Group AB
d) Group O

20. Immunity acquired through vaccination is known as:

a) Natural passive immunity
b) Artificial passive immunity
c) Natural active immunity
d) Artificial active immunity

21. An enzyme in the duodenum that acts on fats is:

a) Pancreatic lipase
b) Pancreatic amylase
c) Trypsinogen
d) Enterokinase

22. The structure which prevents food from entering the wind pipe (trachea) is the:

a) Epiglottis
b) Glottis
c) Tongue
d) Rings of cartilage

23. A Centre where old people are kept and taken care of is called:

a) Health care centre
b) Post-natal clinic
c) A geriatric home
d) Antenatal clinic

24. The causative agent of typhoid fever is:

a) Tubercle bacillus
b) Salmonella typhosus
c) Tryponema pallidum
d) Corynebacterium diphtheriae

25. Fraternal twins result from:

a) The release of two ova
b) One fertilized ovum

c) Cross fertilization
d) Failure of zygote to divide

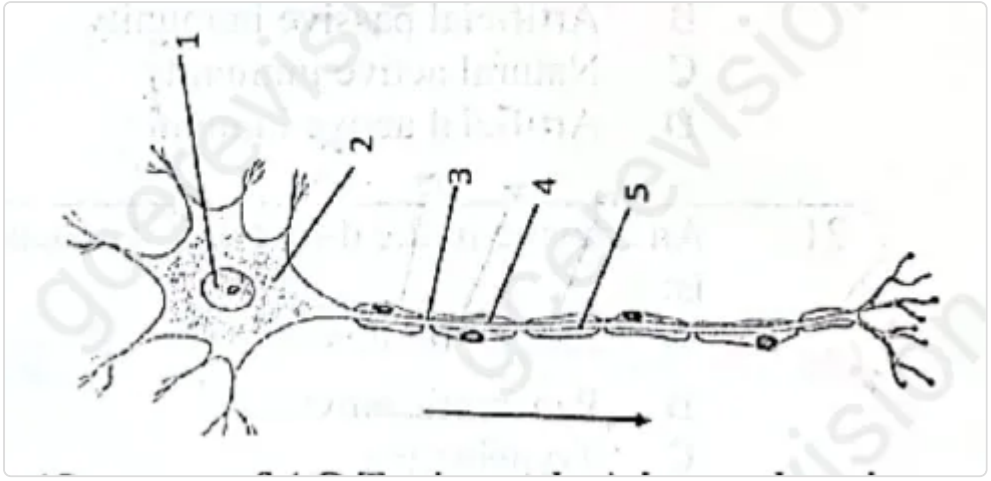
26. The process by which the liver converts excess amino acids into urea is called:

a) Transamination
b) Desaturation
c) Detoxification
d) Deamination

27. Which component of cigarette smoke causes cancer?

a) Smoke
b) Tar
c) Nicotine
d) Carbon monoxide

28. The diagram below is that of the motor neurone.



(Courtsey of AG Toole *et al*., Advance Level Biology Course Companion)

Which numbered structure carries impulses towards the muscle?

Options as printed: A) 1 B) 3 C) 4 D) 5

29. The monthly process in humans that releases an egg from the ovary is called:

a) Ovulation
b) Menstruation
c) Implantation
d) Gestation

30. Which of these blood components is biconcave in shape and lacks a nucleus?

a) thrombocytes
b) Lymphocytes
c) Erythrocytes
d) Monocytes

31. The structures used by an adult tapeworm to attach itself onto the walls of the host's small intestine are the:

Options as printed: A) Rostellum and suckers B) Hooks and suckers C) Proglottids and hooks D) Hooks and rostellum

32. In the carbon cycle the process that converts carbon from fossil fuel to carbon dioxide is:

a) Respiration
b) Decomposition
c) Combustion
d) Fossilization

33. At what stage of the purification process of water is alum or iron III chloride added to water?

a) Sedimentation
b) Screening
c) Filtration
d) Chlorination

34. The hormone responsible for maintaining the lining of the uterus during pregnancy is called:

a) Luteinizing hormone
b) Follicle stimulating hormone
c) Progesterone
d) Prolactin

35. The term homeostasis refers to:

a) The maintenance of a constant body temperature
b) The maintenance of a constant composition of blood and tissue fluid
c) The maintenance of the water content of the body fluids
d) The maintenance of a constant external environment

36. Choose the structure of the skin that is sensitive to heat and cold.

a) Hair plexus
b) Meissner's corpuscle
c) Free nerve endings
d) Pacinian corpuscle

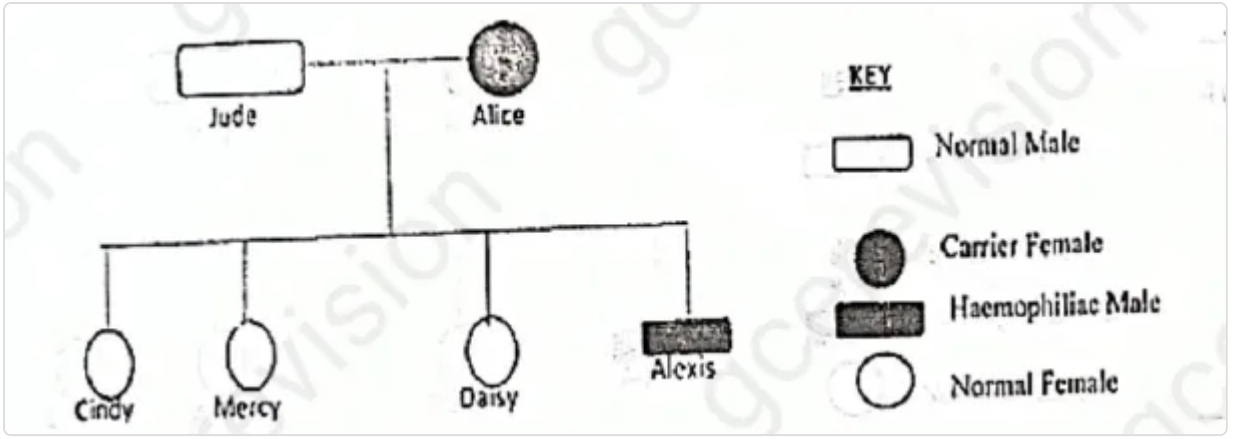
37. Which of the following occurs during Expiration?

a) The intercostal muscles relax
b) The intercostal muscle contract
c) The volume of the thorax increases
d) The volume of the thorax remains the same

38. Identify the blood vessels below that have valves in them.

a) Capillaries
b) Arterioles
c) Veins
d) Venules

39. The pedigree below illustrates the sex-linked recessive trait known as haemophilia in a family.



What is the genotype of Alice?

a) $X^H X^H$
b) $X^H Y$
c) $X^h Y$
d) $X^H X^h$

40. A disease condition in which the body does not have enough red blood cells is:

a) Haemophilia
b) Anaemia
c) Artheriosclerosis
d) Thrombosis

41. A cold compress (ice pack) is applied on the affected joint of a victim of a dislocation as a first aid measure to:

a) Keep bones in position
b) Prevent bleeding
c) Prevent tearing of muscles
d) Reduce pain and inflammations

42. Which enzymes are contained in intestinal juice?

a) Ptyalin, trypsin and lipase
b) Maltase, lipase and erepsin
c) Ptyalin, lipase and pepsin
d) Erepsin, sucrase and ptyalin

43. Identify one cause of miscarriage in human females.

Options as printed: A) Consumption of prescribed drugs B) Insufficient secretion of oestrogen C) Excessive cigarette smoking D) Insufficient secretin of progesterone

44. A disease which spreads worldwide affecting many people at a time is known as a/an:

a) Pandemic
b) Endemic
c) Sporadic
d) Epidemic

45. What determines sex in humans?

a) Alleles
b) Genes
c) Autosomes
d) Heterosomes

46. Identify one hazard of ultraviolet radiation in humans.

a) Obesity
b) Miscarriage
c) Deafness
d) Skin cancer

47. Which class of food is responsible for growth and repair of body tissues?

a) Proteins
b) Vitamins
c) Minerals
d) Carbohydrates

48. In conditions of high salt concentration in the blood:

a) No ADH is secreted by the pituitary gland
b) ADH is secreted by the pituitary gland
c) More salts are absorbed along the nephron
d) Less water is absorbed along the nephron

49. An important rule of good hygiene is to:

a) Exercise the body once in a while
b) Bathe immediately after exercising the body
c) Trim finger and toe nails regularly
d) Taking drugs regularly

50. A benefit of a playground to the community is:

a) A place for the perpetuation of violence
b) A place for the training and practice of games
c) A place for the organizing of religious crusades
d) A place for the lodging of drug addicts

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